INSIDE THE CAVES

by dave of the caves

> The CAVES family of games so far consists of CAVES1, CAVES2, CAVES3, PUBLIC CAVES.
>
> You are invited to come inside the CAVES and see how they are constructed.

Remember CAVES1? It appeared in the May 1973 issue of PCC. Here's how it started:

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IMAGINE YOURSELF AN EXPLORER OF THE FAMOUS
DUZZLEDORF CAVES.  YOU'VE BEEN UNDERGROUND
FOR DAYS, TRIPPING THROUGH THE CAVERNS AND
TUNNELS.  UNFORTUNATELY, YOU'RE LOST, AND
YOUR FOOD HAS RUN OUT.

THERE IS ONLY ONE PATH OUT.  SEE IF YOU
CAN FIND IT.
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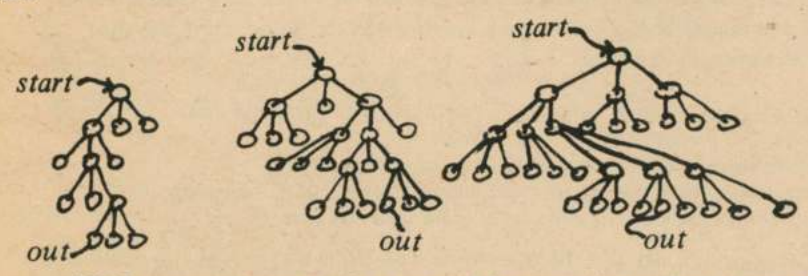
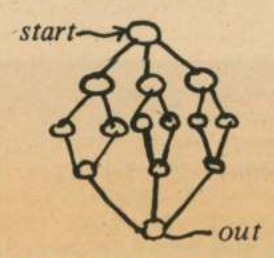
CAVES1 can generate three kinds of CAVES. Here they are –

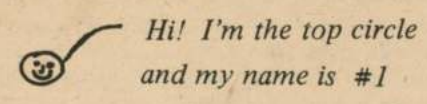


All three start you at the top and let you try to find the only path that leads out. A CAVE could also look like this –

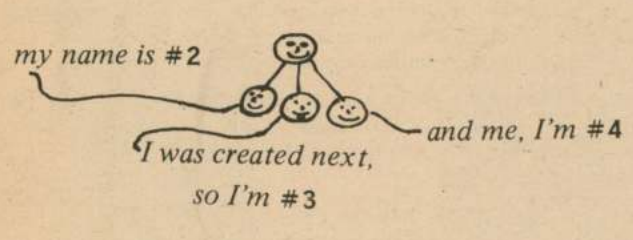


Every path leads out. Not too interesting for a game, is it?

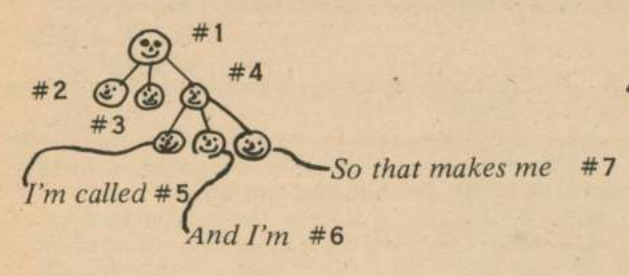
Each of those maps can be built by a CAVES program. First, you have to start –



then you decide how many tunnels (three is my favorite number) and give each a # name –



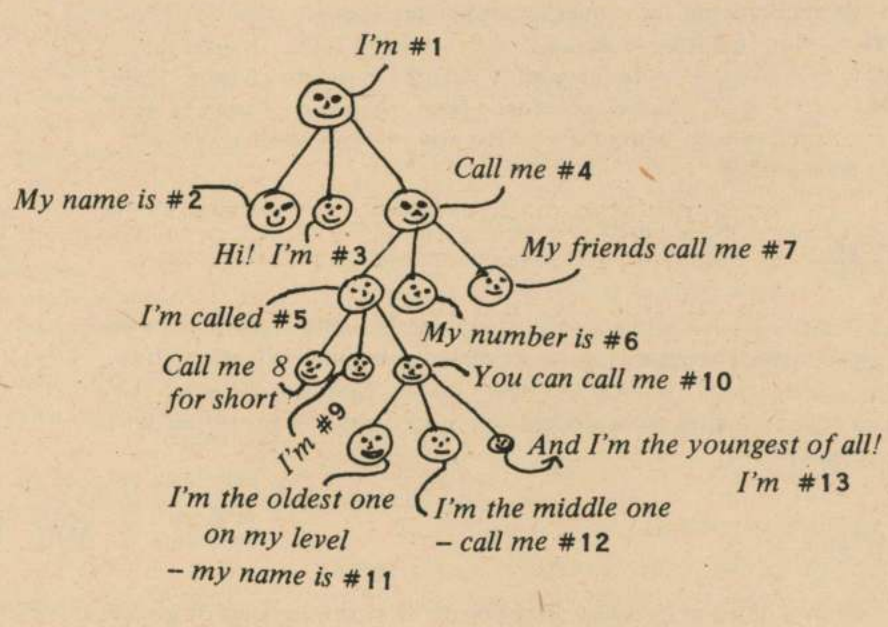
Pick one of the NODES on the second level – (I picked #4 – if you didn't, then try again) – and do the same thing –



A good idea for names is to give each NODE a different one. I like to choose each new name by adding 1 to the last name I gave out.

If people had numbers for names, what would YOUR number be?

So here's the final picture:



So there you are. A little family of connected circles (folks in computer science call them NODES), each with its own name.

Here are the ideas that made this sctructure –

1. Start with the top NODE. Call her #1
2. Do this 4 times –
   - 2A. Do this 3 times –
     - 2A1. Add 1 to the last name
     - 2A2. Make a new tunnel and a new NODE using the new name
   - 2B. Pick one at random
   - 2C. Move down to it
3. Pick the NODE you're in and call it the "out" NODE.

So that is what we'd like to happen. In the CAVES family, tunnels and NODES are created and kept track of by a special set of subroutines called TREE SUBROUTINES. The word TREE is used in computer talk to mean all the different kinds of structures made of tunnels and NODES.

NEXT ISSUE we'll talk about the TREE SUBROUTINES and how you can use them in your own games. Maybe we'll even have time to talk about how they do their jobs!!!